

Return-Decay Residuals and Tail-Risk Forecasting: Timing Artifacts, Conditional Inference, and Cross- Market Evidence

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Three quantities should stay separate

A curve, a performance statistic, and an economic mechanism answer different questions.

Economic edge

The advantage from a trading opportunity. A stylized competition model can motivate decline, but returns alone do not identify participation.

Rolling Sharpe

Average recent return divided by recent volatility. Adjacent windows reuse nearly all their data.

Decay residual

Expected rolling performance minus observed rolling performance. It is a constructed feature, not a direct crowding measurement.

- A useful feature does not automatically establish its proposed economic explanation.

The timeline determines whether this is a forecast

Use month t to predict month $t + 1$.

C_t = curve prediction from earlier months – observed rolling Sharpe at t

- C_t : shortfall relative to an inverse-linear curve.
- The curve uses Sharpe observations ending before t .
- Observed Sharpe includes return t .
- The tail event is return $t + 1$ below a fixed historical threshold.

Strong-looking association can arise without predictability

Simplified Gaussian iid example: zero benchmark; high state means negative rolling Sharpe.

Window	Same-month risk ratio	Next-month population ratio
12 months	2.335	1.000
36 months	1.606	1.000
60 months	1.440	1.000

- Same-month association appears without predictable future returns.
- This isolates timing; it does not reproduce the fitted nonzero curve.
- For 36-month rolling means, adjacent correlation is $35/36 \approx 0.972$.

Test the residual against matched alternatives

Same outcomes and evaluation dates; different representations of past returns.

Forecast design

Separate ridge-logistic models predict each factor's next-month tail event. Thresholds, scaling, and penalty selection precede evaluation.

Comparison models

Historical event rate; return and volatility controls; controls plus residual; controls plus raw Sharpe; constant or linear residual alternatives.

Geographic extension

Six factors in Europe, Japan, and Asia Pacific excluding Japan. Train 2000–2009, validate 2010–2014, evaluate 2015–2024 and 2025–August 2026.

- Regional frozen and annual-refit protocols were fixed before their forecast outcomes were computed.

Regional log loss worsens in the main era

2015–2024 pooled regional forecasts; lower log loss is better.

Model	Frozen fit	Annual refit
Controls	0.30419	0.29941
Controls + hyperbolic residual	0.31123	0.30174
Controls + raw Sharpe	0.30228	0.29879
Controls + constant residual	0.30137	0.29826

- Residual minus controls: +0.00704 frozen; +0.00233 annual.
- Frozen Brier score improves slightly: 0.08673 → 0.08608.

The uncertainty still permits useful gains

Residual-minus-controls log loss; negative values favor the residual.

2015–2024 pooled fit	Observed difference	95% simultaneous interval
Frozen	+0.00704	[−0.03003, +0.06172]
Annual	+0.00233	[−0.01830, +0.03229]

- All 48 final intervals contain zero.
- The target is a running conditional expected score difference.
- No directional conclusion; no established practical equivalence.

Reasonable design choices can reverse the ranking

Post-results US diagnostic: the target definition and feature history interact.

Threshold reference	Feature history	Residual – controls log loss
Full historical reference	Full	−0.00561
Full historical reference	Short	+0.03999
Short historical reference	Full	−0.01384
Short historical reference	Short	−0.04739

- Full feature history starts July 1963; short starts December 1990.
- Threshold changes redefine events; rows across thresholds do not share identical labels.
- Penalty selection also contributes; this diagnostic is descriptive.

A better representation is not new economic information

A feature computed from X cannot add information conditional on all of X .

If C_t is computed from X_t , conditioning on both gives the same information as conditioning on X_t .

- X includes all construction inputs, fitted history, and time; a simple learner may still benefit from the feature.
- Equivalent feature bases can change ridge penalties and fitted probabilities.
- Large gains over a weak comparator can disappear against raw Sharpe.
- Independent holdings or flow evidence is needed for a crowding claim.

Requirements for a defensible forecast claim

Timing, benchmarks, and uncertainty determine what the evidence supports.

- Separate current-loss description from next-month prediction.
- Compare matched representations, histories, targets, and tuning rules.
- Report scoring-rule reversals and unfavorable outcomes.
- Treat crowding and trading value as questions still requiring independent evidence.

Source: github.com/ChorokLeeDev/factor-regime-revision